SRM INSTITUTE OF SCIENCE & TECHNOLOGY

PASS SCIENCE WORKS

R&D Internship Evaluation

Model : Single Neuron Tanh Regression | Target $x = 1.0$, $y = 2.0$

1. Model Definition & Initial Conditions :

Architecture & Equations :

- Pre-activation : $z = w \cdot x + b$
- Hidden activation : $h = \tanh(z) = (e^{z} - e^{-z}) / (e^{z} + e^{-z})$
- Loss function (Squared error) : $J = 1/2(\hat{y} - y)^2$

Initial Values (Step $t = 0$) :

- Input : $x = 1.0$ | Target : $y = 2.0$
- Input weight : $w_0 = 0.5$ | Hidden bias : $b_0 = 0.0$
- Output weight : $v_0 = 1.0$ | Output bias : $c_0 = 0.0$
- Learning rate : $\eta = 0.1$ (simultaneous updates)

2. Forward Pass : Step by Step Numerical Evaluation (Step 0)

Quantity	Formula & Substitution	Exact Computed value
Pre activation z	$z = (0.5)(1.0) + 0.0$	0.5000000000
Hidden Activation h	$h = \tanh(0.5) = \frac{(e^{0.5} - e^{-0.5})}{(e^{0.5} + e^{-0.5})}$	0.4621171573
Prediction $\hat{y}$	$\hat{y} = (1.0)(0.4621171573) + 0.0$	0.4621171573
Prediction error δ	$\delta = \hat{y} - y = 0.4621171573 - 2.0$	-1.5378828427
Initial loss J_0	$J = 1/2\ (-1.5378828427)^2$ $= 1/2(2.3650836379)$	1.1825418190

3. Backward Pass: Complete Chain-Rule Derivations:

Applying the multivariable chain rule from the scalar loss J backward through $\hat{y}$, h, z:

*Loss Derivation w.r.t Output Prediction:

$$\partial J/\partial \hat{y} = \partial[1/2(\hat{y}-y)^2]/\partial \hat{y} = (\hat{y}-y) = \delta = -1.537888428427$$

* Gradient w.r.t Output Bias c:

$$\partial J/\partial c = (\partial J/\partial \hat{y}) \cdot (\partial \hat{y}/\partial c)$$

Since $\partial \hat{y}/\partial c = 1$:

$$\partial J/\partial c = \delta \cdot 1 = -1.537888428427$$

* Gradient w.r.t Output weight v:

$$\partial J/\partial v = (\partial J/\partial \hat{y}) \cdot (\partial \hat{y}/\partial v)$$

Since $\partial \hat{y}/\partial v = h$:

$$\partial J/\partial v = \delta \cdot h = (-1.537888428427)(0.4621171573)$$

$$\partial J/\partial v = -0.7106820475$$

* Backpropagation into Hidden Layer:

By chain rule: $\partial J/\partial z = (\partial J/\partial \hat{y}) \cdot (\partial \hat{y}/\partial h) \cdot (\partial h/\partial z)$

$\cdot \partial \hat{y}/\partial h = v$

· Derivative of tanh: $dh/dz = 1 - \tanh^2(z) = 1 - h^2$

$$= 1 - (0.4621171573)^2 = 1 - 0.2135522672$$

$$= 0.7864477328$$

$\therefore \partial J/\partial z = \delta \cdot v \cdot (1-h^2) = (-1.537888428427)(1.0)(0.7864477328)$

$$= -1.2094644752$$

* Gradient w.r.t Input weight w:

$\partial J/\partial w = (\partial J/\partial z) \cdot (\partial z/\partial w) = (\partial J/\partial z) \cdot x$

Since $x = 1.0$:

$\partial J/\partial w = (-1.209464475 2)(1.0) = -1.209464475 2$

* Gradient w.r.t Hidden Bias b:

$\partial J/\partial b = (\partial J/\partial z) \cdot (\partial z/\partial b) = (\partial J/\partial z) \cdot 1$

Since $\partial z/\partial b = 1$:

$\partial J/\partial b = (-1.209464475 2)(1.0) = -1.209464475 2$

4. Central Finite Difference Numerical Gradient Check ($\varepsilon = 10^{-6}$)

Central finite difference formula : $[\partial J/\partial \theta]_{num}$

$= [J(\theta+\varepsilon) - J(\theta-\varepsilon)]/(2\varepsilon)$

with $\varepsilon = 1.0 \times 10^{-6}$

Parameter θ	Analytical ($\partial J/\partial \theta$)	Numerical	Absolute	Tolerance
Input Weight w	$-1.209464475 2$	$-1.209464475 1$	1.13×10^{-10}	Pass ($<<10^{-5}$)
Hidden Bias b	$-1.209464475 2$	$-1.209464475 1$	1.13×10^{-10}	Pass ($<<10^{-5}$)
Output Weight v	$-0.T106820475$	$-0.T106820475$	5.25×10^{-11}	Pass ($<<10^{-5}$)
Output Bias c	$-1.53788284 27$	$-1.53788284 26$	1.12×10^{-10}	Pass ($<<10^{-5}$)

5. Simultaneous Parameter Update Rule (Step $t=0 \to t=1$)

Gradient descent update: $\theta_{t+1} = \theta_t - \eta \cdot (\partial J/\partial \theta_t)$ with learning rate $\eta = 0.1$. All parameters are updated simultaneously using gradientients evaluated at original parameters θ_0:

$w_1 = w_0 - \eta\ (\partial J/\partial w) = 0.5 - (0.1)(-1.2094644452)$

$w_1 = 0.5 + 0.120946447 5 = 0.6209464475$

$v_1 = v_0 - \eta\ (\partial J/\partial v) = 1.0 - (0.1)(-0.7106820475)$

$v_1 = 1.0 + 0.07106820848 = 1.07106820847$

$b_1 = b_0 - \eta\ (\partial J/\partial b) = 0.0 - (0.1)(-1.2094644752)$

$b_1 = 0.0 + 0.120946447 5 = 0.1209464475$

$c_1 = c_0 - \eta\ (\partial J/\partial c) = 0.0 - (0.1)(-1.5378828427)$

$c_1 = 0.0 + 0.1537882843 = 0.1537882843$

6. Physical Meaning of Negative Gradient Direction:

At initialization, the network outputs $\hat{y} = 0.4621$, which is significantly smaller than the target $y = 2.0$. The prediction error is $\delta = \hat{y} - y = -1.5379 < 0$. Since the loss is $J = 1/2\ (\hat{y} - y)^2$, the slope $\partial J/\partial \hat{y}$ is negative. Increasing w, b, v & c increases $\hat{y}$ because gradient descent steps in the opposite direction of the gradient ($-\nabla J$), the negative sign cancel: $\Delta\theta = -\eta(\partial J/\partial \theta) >0$